

This Zenodo deposit contains a publicly available description of the Dataset:

"The Parkinson's disease kinase LRRK2 impairs release sites of vulnerable dopamine axons - APEX2 proteome analysis".

Dataset Description:

We and others previously reported that knock-in mice expressing the pathogenic hyperactive Lrrk2G2019S mutation, exhibit deficits in dopamine release within the striatum. To investigate the underlying molecular mechanisms, we conducted axonal proteome analysis of a subcluster within vulnerable Aldh1a1+ dopamine axons, defined by Annexin 1 (Anxa1+), confirmed subtle proteomic changes across comparisons.

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ASAP Team: Dombeck

Dataset Name: dombeck-mouse-ms-p-striatum-nigra-apex2, v0.1

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ASAP Lab:

ASAP Project: Redefining PD pathophysiology mechanisms in the context of heterogeneous substantia nigra neuron subtypes.

Project Description: The degeneration of substantia nigra dopamine neurons leads to deficits in striatal dopaminergic transmission that underlie the motor symptoms of Parkinson's disease. We have revealed molecularly distinct, pro-motor and anti-motor dopamine subtypes. We hypothesize that motor dysfunction in Parkinson's Disease is caused by an imbalance in pro- and antimotor dopamine signaling.

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This dataset is made available to researchers via the ASAP CRN Cloud: cloud.parkinsonsroadmap.org. Instructions for how to request access can be found in the [User Manual](#).

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This Zenodo deposit was created by the ASAP CRN Cloud staff on behalf of the dataset authors. It provides a citable reference for a CRN Cloud Dataset